Trainer Setup Guide for Lab 2: Storage and Catalog Access

The metastore has no storage root, so learners must create catalogs with a MANAGED LOCATION. This guide sets up that storage once, gives learners the privileges they need, and tests it. Allow 30 to 45 minutes.

What you will set up

Item	Name used in this guide
ADLS Gen2 storage account	<storage-account> (lowercase, globally unique)
Container	uc-lab
Access Connector for Azure Databricks	uc-lab-connector
Storage credential (Unity Catalog)	uc_lab_cred
External location (Unity Catalog)	uc_lab_location
Learner group (Entra ID)	uc-lab-learners

Region: create every Azure resource in the same region as the Databricks workspace and in the same resource group.

Part A: Azure storage

A1. Create the storage account

1. In the Azure portal, go to Storage accounts > Create.
2. Basics: choose the subscription, resource group and region of the workspace. Enter the storage account name. Performance: Standard. Redundancy: LRS.
3. Advanced tab: tick Enable hierarchical namespace. This is what makes it ADLS Gen2.
4. Networking tab: keep Enable public access from all networks for the lab.
5. Review + create > Create.

A2. Create the container

1. Open the storage account > **Data storage** > **Containers** > **+ Container**.
2. Name: uc-lab. Create.

A3. Create the Access Connector

1. Search the portal for **Access Connector for Azure Databricks** > **Create**.
2. Same resource group and region. Name: uc-lab-connector. Keep the system-assigned managed identity. Create.
3. Open the connector and copy its **Resource ID** from the Overview page (it starts with /subscriptions/...).

A4. Let the connector use the storage

1. Open the storage account > **Access control (IAM)** > **Add** > **Add role assignment**.
2. Role: **Storage Blob Data Contributor**.
3. Members: **Managed identity** > **Select members** > Access Connector for Azure Databricks > uc-lab-connector.
4. **Review + assign**. The role can take a few minutes to take effect.

> **Check:** The storage account's IAM > Role assignments shows uc-lab-connector as Storage Blob Data Contributor.

Part B: Unity Catalog storage credential and external location

Do this in the workspace as your admin account. Use the shared serverless SQL warehouse for the SQL.

B1. Make sure you can create them

If step B2 or B3 gives a permission error, run this first (account admins can grant metastore privileges):

```
GRANT CREATE STORAGE CREDENTIAL, CREATE EXTERNAL LOCATION
 ON METASTORE TO `saurabh@uptut.com`;
```

B2. Create the storage credential

1. Click **Catalog** > **External data** > **Credentials** > **Create credential**.
2. Credential type: **Azure Managed Identity**. Name: uc_lab_cred.
3. Access connector ID: paste the Resource ID from A3. Create.

B3. Create the external location

```
CREATE EXTERNAL LOCATION IF NOT EXISTS uc_lab_location
  URL 'abfss://uc-lab@<storage-account>.dfs.core.windows.net/'
  WITH (STORAGE CREDENTIAL uc_lab_cred)
  COMMENT 'Storage for Lab 2 learner catalogs';
```

1. In Catalog > External data > External locations, open uc_lab_location and click Test connection.

> Check: All tests pass. If Read or Write fails, wait a few minutes for the role from A4 and test again.

Part C: Learner privileges

1. In Entra ID, create a security group uc-lab-learners and add all six learner accounts (Andrew, Bob, Carol, Dave, Saurabh Learner, Yu).
2. In the account console, go to User management > Groups > Add group and add uc-lab-learners.
3. Run:

```
GRANT CREATE CATALOG ON METASTORE TO `uc-lab-learners`;
GRANT CREATE MANAGED STORAGE ON EXTERNAL LOCATION uc_lab_location
  TO `uc-lab-learners`;

SHOW GRANTS `uc-lab-learners` ON METASTORE;
SHOW GRANTS ON EXTERNAL LOCATION uc_lab_location;
```

> Check: The group holds CREATE CATALOG on the metastore and CREATE MANAGED STORAGE on uc_lab_location.

Part D: Test as one learner

1. Sign in to the workspace as one learner account (for example Bob Learner).
2. In SQL Editor on the shared warehouse, run:

```
CREATE CATALOG IF NOT EXISTS precheck_bob
  MANAGED LOCATION 'abfss://uc-lab@<storage-account>.dfs.core.windows.net/precheck_bob';

CREATE SCHEMA precheck_bob.s1;
CREATE TABLE precheck_bob.s1.t1 AS SELECT 1 AS id;
SELECT * FROM precheck_bob.s1.t1;

DROP CATALOG precheck_bob CASCADE;
```

Result	What to do
All statements succeed	Ready for Lab 2.
No CREATE CATALOG	Check Part C, step 3.
No CREATE MANAGED STORAGE	Check the second GRANT in Part C.
Access denied to storage	Check the role in A4 and the Test connection in B3.

Before the session

- Give learners the storage account name so they can complete the path abfss://uc-lab@<storage-account>.dfs.core.windows.net/<prefix>.
- The shared SQL warehouse is serverless and learners can manage its permissions.
- Every learner completed Lab 1 and customers.csv is ready to share.

After the session (optional clean-up)

```
DROP CATALOG IF EXISTS <prefix>_catalog CASCADE;   -- one per learner
REVOKE CREATE CATALOG ON METASTORE FROM `uc-lab-learners`;
REVOKE CREATE MANAGED STORAGE ON EXTERNAL LOCATION uc_lab_location
  FROM `uc-lab-learners`;
```

Keep the storage account, connector, credential and external location if you plan to run the lab again.